

# EMS Dashboard

Connected

Supervisory Python EMS with six scenarios, manual scenario control, manual price input, virtual battery charge, hydrogen estimation and live INA measurements in V, mA and mW.

## Demo and EMS control

EMS: started

Start the EMS system first. Then start the demo to reset the 96-slot timeline, begin CSV logging and activate the variable load trigger. Raw logging can run independently of the demo.

Bridge: OK

Reset

Start EMS system

Stop EMS system

Start demo + logging

Stop demo

Auto mode

Manual mode

Start raw log

|                    |                      |
|--------------------|----------------------|
| Raw log<br>Stopped | Raw log file<br>-    |
| Raw samples<br>0   | Last raw update<br>- |

Price test date

07 / 05 / 2026

Load prices for date

Manual electricity price

0.97727

Apply manual price

Hydrogen estimation (%)

22.1

Initiate H2 estimation manually

Virtual battery charge state (%)

10.0

Apply virtual battery state

Control mode

manual

Demo status

Stopped

Current slot

95

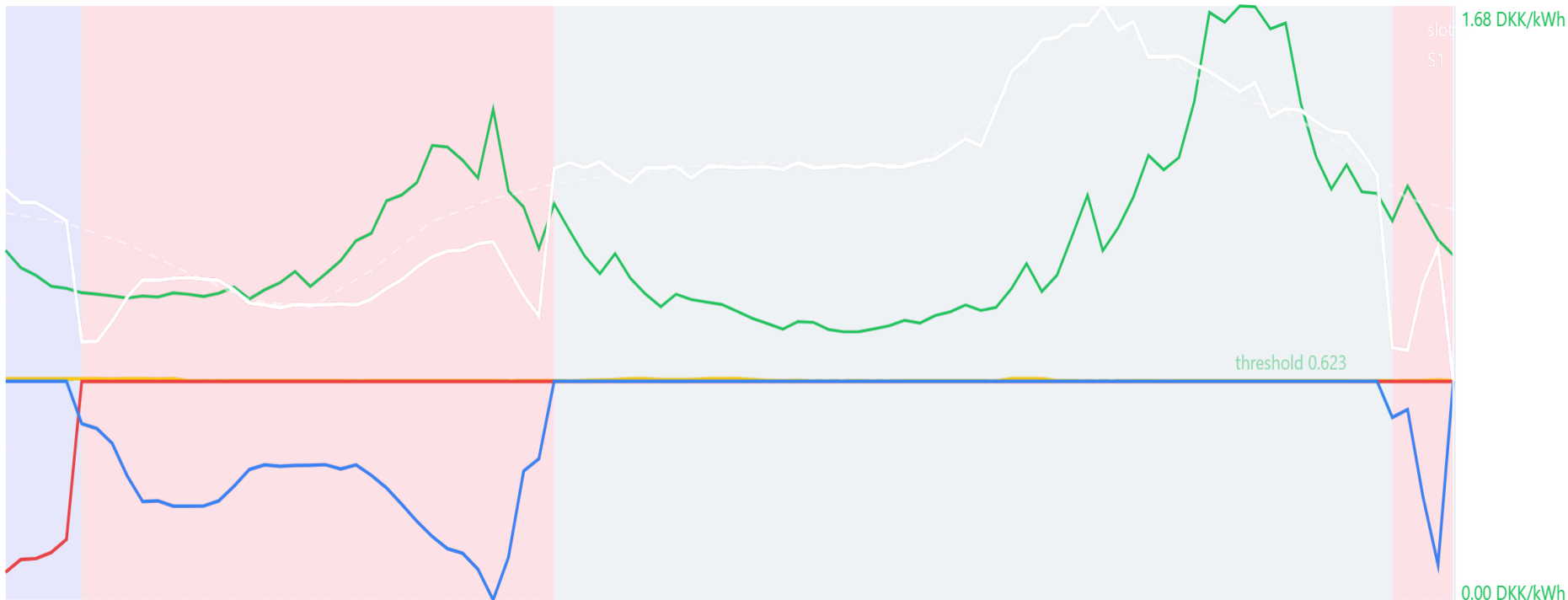
Scenario

S1

System state timeline

One plot showing the selected price curve, demand profile and measured INA power in mW for PV, load, battery and PEM. Scenario background is shown where a slot has been logged.

- Price
- Price threshold
- Demand
- PV power
- Load power
- Battery power
- PEM power



Time  
23:45

Interval  
23:45-00:00

Price  
0.97727 DKK/kWh

Demand  
35.0 mW

PV power  
0.1 mW

Load power  
-0.0 mW

Battery power  
**-0.1 mW**

PEM power  
**-0.0 mW**

Virtual battery charge  
**10.0 %**

Hydrogen estimation  
**22.1 %**

Actual scenario  
**S1**

Current EMS decision

One 96-slot demo cycle was completed. Demo logging was stopped, the variable load trigger was disabled, and the system was returned to safe S1.

EMS system      Started

Target scenario    S1

Actual scenario    S1

Reject reason

Price state        HIGH

Current price      0.97727 DKK/kWh

Manual scenarios

Selecting a scenario switches the EMS to manual mode and sends the scenario once.

S1 Grid → Load

S2 Grid → Load, PV → Battery

S3 Grid → Load, PV → PEM

S4 PV → Load

S5 Battery → Load

S6 PEM → Load

|                 |                                                   |
|-----------------|---------------------------------------------------|
| Current demand  | 35.0 mW                                           |
| PV available    | No                                                |
| Log file        | /app/python/logs/ems_demo_20260512_142228.csv     |
| Last log update | Demo cycle completed at 2026-05-12T14:46:29+02:00 |
| Error           | -                                                 |

Storage states

|                                   |                         |
|-----------------------------------|-------------------------|
| Current battery voltage           | 3.693 V                 |
| Initial SoC lookup voltage        | 3.693 V                 |
| Lookup-estimated real battery SoC | 4.7 %                   |
| Integrated real battery SoC       | 4.7 %                   |
| Integrated real battery charge    | 93.136 mAh / 2000.0 mAh |
| Virtual battery SoC               | 10.0 %                  |
| Virtual battery capacity          | 10.00 mAh               |
| Virtual battery charge            | 0.999 mAh               |
| Virtual battery state             | empty                   |

INA sensor measurements

|                 |          |
|-----------------|----------|
| PV voltage      | 2.982 V  |
| Battery voltage | 3.693 V  |
| PEM voltage     | 0.883 V  |
| Load voltage    | 5.123 V  |
| PV current      | 0.03 mA  |
| Battery current | -0.03 mA |
| PEM current     | -0.04 mA |
| Load current    | -0.01 mA |
| PV power        | 0.07 mW  |

|                     |         |
|---------------------|---------|
| Hydrogen volume     | 9.37 mL |
| Usable hydrogen     | 1.77 mL |
| Hydrogen estimation | 22.1 %  |
| Hydrogen mode       | idle    |

|               |                 |
|---------------|-----------------|
| Battery power | -0.09 mW        |
| PEM power     | -0.04 mW        |
| Load power    | -0.04 mW        |
| Load trigger  | LOW             |
| Arduino mode  | S1 Grid -> Load |